Alvaro Carballeira

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Education

Brown University. Providence, RI, USA.
PhD in Mathematics. *Sep 2024 – Present*

- **Advisor:** Javier Gómez-Serrano.
- **Research interests:** Partial Differential Equations, Computer-assisted Proofs, Nonlinear Schrodinger Equations, Fluid Mechanics.

Instituto de Ciencias Matemáticas (ICMAT) Madrid, Spain
Severo Ochoa Contract for Students of a Master Degree *Sep 2023 – Aug 2024*

- The program consists on the collaboration of students with ICMAT researchers. I worked with María del Mar González (professor at UAM) and Marco Antonio Fontelos (researcher at ICMAT) in the field of free boundary problems. More specifically, we targeted problems such as the mean curvature flow of a moving boundary, the Stefan problem or the flow of a binary fluid using phase field methods.

Universidad Autónoma de Madrid. Madrid, Spain
Masters Degree in Mathematics and Applications. *Sep 2023 – Jun 2024*

- **Master thesis:** The Cahn-Hilliard Equation. Phase Field Methods for Moving Boundary Problems.
- **Average grade:** 9.95/10.00

Universidad Autónoma de Madrid. Madrid, Spain
B.Sc. in Mathematics. *Sep 2019 – Jun 2023*

- **Bachelor thesis:** Neural Networks. Applications to numerical aproximation of PDEs.
- **Average grade:** 9.02/10.00

Publications and preprints

- *Blow-up for energy supercritical defocusing nonlinear Schrödinger equations in dimensions three, four and five* (with K. Buck, J. Gómez-Serrano, and J. Shi). Preprint, 2026. arXiv:2609:23685.
- *The energy of fractional Allen–Cahn layers in dimension one* (with D. Galant, J. Gómez-Serrano). Preprint, 2026. arXiv:2608.06226.

Awards and Achivements.

Department of Mathematics, Universidad Autonoma de Madrid Madrid, Spain
Scholarship for the Master's Degree in Mathematics and Applications. *Jun 2023*

- I was awarded one of the three scholarships offered by the Department of Mathematics to cover the tuition fees for the Master's Degree in Mathematics and Applications.

Talks and presentations

- *Blow-up for energy supercritical, defocusing NLS. Parts I and II.* Graduate Lecture Series in Analysis and PDEs (GLESPA), Brown University. September 29 and October 6, 2026.

Attended workshops and conferences.

- *Dynamics, Topology and Computations.* Jagiellonian University in Kraków. June 2025.
- *Particles, Fluids and Patterns: Analytical and Computational Challenges.* GSSI (L'Aquila, Italy). June 2025
- *Riviere-Fabes Symposium.* University of Minnesota. April 2025.
- *Conference on fluids and computer assisted proofs.* Princeton University. March 2025.
- *Computer-assisted proof in nonlinear analysis.* CRM (Montreal, Canada). September 2024.
- *European Congress of Mathematics.* Universidad de Sevilla (Sevilla, Spain). July 2024.
- *JAE School of Mathematics.* ICMAT (Madrid, Spain). June-July 2024.
- *2023 Thematic Period on PDEs.* ICMAT-UAM (Madrid, Spain) September-December 2023.
- *JAE School of Mathematics.* ICMAT (Madrid, Spain). June-July 2023.

Teaching

Introduction to Calculus, Part I – Math 90 Fall 2026
Lecturer/Instructor *Brown University*

Introduction to Calculus, Part II – Math 100 Fall 2025
Teaching assistant *Brown University*

Directed Reading Program Fall 2025
Mentor *Brown University*

Service and outreach activities

- Co-organizer of the Brown University Graduate Student Seminar.
- Co-organizer of the AMS New England Graduate Student Conference.
- Referee for the Proceedings of the AMS.

Certificates

Cambridge Assestment English: Certificate in Advanced English *Sep 2021*
Overall score: 203. Common European Framework level: C2.

IELTS Academmic Text *Dec 2023*
Overall band score: 8.5.